

SentinelAI – AI Powered Digital Public Safety Intelligence Platform

Economic Times AI Hackathon 2026 – Project Blueprint

Vision

SentinelAI is an AI-powered Digital Public Safety Intelligence Platform designed to proactively detect, understand, predict, and prevent digital fraud before citizens become victims. It targets Digital Arrest, Banking Fraud, KYC scams, Phishing, Investment Fraud, Remote Access scams, and emerging scam variants.

Core Innovation

Instead of relying on static keyword detection, SentinelAI converts each analyzed scam into a Behavioral Fingerprint representing authority impersonation, psychological manipulation, communication style, payment methods, technology used, execution pattern, and victim profile. New incidents are compared against historical fingerprints to identify known scams, evolving variants, or potentially new campaigns.

Stakeholders

- Citizens – Real-time scam analysis and protection
- Law Enforcement – Fraud intelligence, hotspot analysis, connected fraud campaigns
- Banks – Fraud intelligence APIs and scam indicators
- Telecom Providers – Scam call intelligence and alerts
- Government Agencies – National threat intelligence and trend analysis

High-Level Architecture

Threat Sources (Text, URL, PDF, Image, Transcript) → Threat Intelligence Core → Feature Extraction → LLM Threat Intelligence (Gemini) → Behavioral Fingerprint Engine → Adaptive Scam Evolution Engine → Citizen Protection Engine → Fraud Knowledge Graph (Neo4j) → Threat Intelligence APIs → Consumers (Citizen App, Police Dashboard, Banks, Telecom).

Backend Modules

1. Threat Intelligence Core – Ingest PDFs, URLs, screenshots, OCR, complaints.
2. Threat Feature Extraction – Extract phone numbers, URLs, UPI IDs, entities, scam indicators.
3. LLM Threat Intelligence – Produce structured cyber threat intelligence using Gemini.
4. Behavioral Fingerprint Engine – Build scam DNA from authority, psychology, technology, payment and execution patterns.
5. Adaptive Scam Evolution Engine – Detect known scams, variants and novel scam campaigns.
6. Citizen Protection Engine – Threat score, explainability and actionable recommendations.
7. Fraud Knowledge Graph – Connect scams, victims, UPI IDs, phones, authorities and locations.
8. Threat Intelligence APIs – Serve intelligence to frontend and external systems.

Frontend Pages

Dashboard – Threat overview, trends, heatmap, recent intelligence.

Threat Analyzer – Main demo page supporting text, URL, PDF, image and transcript analysis with Threat Score, Behavioral Fingerprint, Variant Detection, Explainability, Timeline, Citizen Guidance, Evidence Extraction and downloadable report.

Knowledge Graph – Interactive fraud relationship visualization.

Scam Intelligence – Variant timelines and threat feed.

Law Enforcement – Fraud hotspots and campaign intelligence.
System Health – Backend, AI, database and queue status.

Technology Stack

Frontend: React (future Streamlit prototype), Framer Motion, Recharts.
Backend: FastAPI, Python, SQLAlchemy, PostgreSQL, Neo4j, FAISS.
AI: Gemini, Sentence Transformers, EasyOCR, PyMuPDF.
Deployment: Docker, Docker Compose.

Hackathon Demo Flow

1. User submits suspicious content.
2. Threat document created.
3. Features extracted.
4. Gemini generates structured intelligence.
5. Behavioral fingerprint created.
6. Evolution engine detects scam variant.
7. Threat score and explainability generated.
8. Citizen guidance displayed.
9. Knowledge graph updated.
10. Intelligence report downloadable.

Future Scope

Real-time voice analysis, QR code scanning, browser extension, WhatsApp integration, telecom APIs, banking APIs, deepfake voice detection, multilingual support, geospatial crime intelligence and adaptive continual learning.

Value Proposition

SentinelAI is positioned as a Digital Public Safety Intelligence Platform rather than a simple scam detector. Its differentiator is the combination of behavioral fingerprinting, explainable threat intelligence, scam variant detection, and proactive citizen protection.